CAGAYAN STATE UNIVERSITY
Aparri Campus
College of Information and Computing Sciences
2nd Semester, 2025-2026

Term:	*Prelims*	**Date of Examination:**	
Subject Code:	*IT122 COMPUTER PROGRAMING 2*	**Score**	
Student Name:		**Year and Section:**	

PART I – MULTIPLE CHOICE (30 items)

Direction: Choose the correct answer. Write the letter only ***ERASURES OR ANY ITERATIONS MEANS WRONG****.*

1. A system requires fast lookup and insertion of key-value pairs. Which structure is MOST appropriate for this requirement?
   A. Dictionary
   B. List
   C. Array
   D. Stack
2. A developer notices frequent collisions in a hashtable. What is the BEST strategy to improve performance?
   A. Increase hash collisions
   B. Use a better hash function
   C. Remove keys
   D. Convert to array
3. When managing ordered data where duplicates are allowed, which structure is MOST suitable?
   A. Dictionary
   B. HashTable
   C. List
   D. Set
4. If a program requires accessing elements by index frequently, which data structure should be prioritized?
   A. Stack
   B. Queue
   C. Dictionary
   D. List
5. A developer needs to safely retrieve a value without risking an exception if the key does not exist. Which method is BEST?
   A. Add()
   B. Remove()
   C. TryGetValue()
   D. ContainsValue()
6. In a system where memory efficiency is critical, which factor should be analyzed when choosing between List and Dictionary?
   A. Color of variables
   B. Key uniqueness
   C. UI design
   D. Loop structure
7. A programmer needs to iterate through key-value pairs efficiently. Which approach is MOST appropriate?
   A. Using DictionaryEntry
   B. Using string concatenation
   C. Using recursion
   D. Using break statements
8. When comparing Hashtable and Dictionary, which scenario BEST justifies using Dictionary?
   A. When boxing/unboxing should be avoided
   B. When using legacy code
   C. When keys are unknown
   D. When no iteration is needed
9. A system crashes due to null values in string manipulation. What is the BEST preventive approach?
   A. Ignore null values
   B. Use null checking before operations
   C. Remove all strings
   D. Restart program
10. Which scenario BEST demonstrates the use of a List over an Array?
    A. Fixed number of elements
    B. Dynamic resizing required
    C. Constant memory
    D. Static allocation
11. A developer needs to ensure no duplicate keys exist. Which structure enforces this automatically?
    A. List

B. Array
C. Dictionary
D. Stack

12. When removing elements frequently from the middle of a collection, which structure performs BETTER?
A. List
B. Linked List
C. Dictionary
D. Stack

13. A program requires reversing strings efficiently. Which approach is BEST?
A. Manual swapping
B. Using built-in reverse methods
C. Deleting characters
D. Ignoring order

14. Which method ensures safe addition of elements only if the key does not already exist?
A. Add()
B. TryAdd()
C. Insert()
D. Append()

15. A system needs to check if a key exists before performing operations. Which is MOST efficient?
A. ContainsKey()
B. ContainsValue()
C. ToString()
D. Clear()

16. Which structure is BEST suited for implementing a caching system?
A. Stack
B. Queue
C. Dictionary
D. Array

17. When working with large datasets, which factor MOST affects Dictionary performance?
A. Key hashing quality
B. UI design
C. Variable naming
D. Loop type

18. A developer wants to convert all characters in a string to uppercase efficiently. What is BEST?
A. Manual ASCII conversion
B. Built-in string method
C. Loop without condition
D. Delete characters

19. Which situation BEST requires the use of ToString()?
A. Converting objects for display
B. Deleting data
C. Sorting arrays
D. Creating loops

20. A system requires FIFO behavior. Which structure should be used?
A. Stack
B. Dictionary
C. Queue
D. List

21. A developer needs to optimize searching operations. Which approach is BEST?
A. Linear search in list
B. Use Dictionary
C. Nested loops
D. Recursion only

22. Which method is MOST appropriate for clearing all elements in a collection?
A. Remove()
B. DeleteAll()
C. Clear()
D. Reset()

23. When debugging a program, which issue is MOST likely caused by incorrect hashing?
A. UI lag
B. Data collisions
C. Syntax error
D. Memory leak

24. A developer must ensure thread safety while accessing collections. Which should be considered?
A. Use synchronized collections
B. Use more variables
C. Avoid loops
D. Use strings

25. Which operation is MOST efficient in a Dictionary?
A. Sequential search
B. Random access by key
C. Index-based removal
D. Sorting

26. A program frequently concatenates strings. What is the BEST optimization?
A. Use StringBuilder
B. Use List
C. Use Array
D. Use Dictionary

27. Which structure is BEST when order matters and indexing is required?
A. Dictionary

B. List
C. Hashtable
D. Queue

28. A developer encounters boxing issues in Hashtable. What is the BEST alternative?
A. Array
B. List
C. Dictionary
D. Stack

29. Which scenario BEST demonstrates misuse of Dictionary?
A. Using duplicate keys
B. Storing key-value pairs
C. Fast lookup
D. Unique mapping

30. When designing scalable systems, which is MOST important?
A. Data structure efficiency
B. Variable names
C. Comments
D. Output format

31. Which method is BEST for checking value existence in Dictionary?
A. ContainsKey()
B. ContainsValue()
C. TryAdd()
D. Remove()

32. A program must handle frequent insertions and deletions. Which structure is BEST?
A. Array
B. Linked List
C. Dictionary
D. Stack

33. Which is the MOST efficient way to iterate through Dictionary?
A. For loop with index
B. Foreach loop
C. While loop only
D. Recursion

34. A developer needs to prevent runtime errors when accessing missing keys. What is BEST?
A. Use TryGetValue()
B. Use Add()
C. Use Clear()
D. Use Remove()

35. Which structure is BEST for LIFO operations?
A. Queue
B. Stack
C. List
D. Dictionary

36. When comparing performance, which is MOST efficient for lookup?
A. Array
B. List
C. Dictionary
D. Stack

37. A program manipulates strings heavily. What is MOST important?
A. Immutability awareness
B. Variable naming
C. Loop count
D. Output design

38. Which scenario BEST uses List instead of Dictionary?
A. Key-value mapping
B. Ordered collection
C. Unique keys
D. Fast lookup

39. A system experiences slow performance due to resizing. What is BEST solution?
A. Predefine capacity
B. Add more loops
C. Remove methods
D. Change syntax

40. Which method converts data into readable string format?
A. Parse()
B. ToString()
C. Convert()
D. Format()

41. Which structure BEST supports dynamic growth?
A. Array
B. List
C. Stack
D. Queue

42. A developer must optimize memory usage. What should be considered?
A. Data structure choice
B. Comments
C. UI layout
D. Syntax

43. Which issue arises from improper string handling?
A. Memory inefficiency
B. Syntax error
C. Loop failure
D. UI bug

44. Which method is BEST for adding items safely in Dictionary?
    A. Add()
    B. TryAdd()
    C. Insert()
    D. Append()
45. A system requires mapping IDs to objects. Which is BEST?
    A. List
    B. Dictionary
    C. Stack
    D. Queue
46. Which structure is BEST for sequential access only?
    A. Stack
    B. Queue
    C. List
    D. Dictionary
47. Which approach BEST improves code readability?
    A. Proper naming and structure
    B. More loops
    C. More variables
    D. Less spacing
48. A program frequently searches data. What is BEST optimization?
    A. Use hashing
    B. Use loops
    C. Use recursion
    D. Use arrays
49. Which method removes a specific key-value pair?
    A. Delete()
    B. Clear()
    C. Remove()
    D. Erase()
50. Which factor MOST affects scalability of collections?
    A. Data structure design
    B. Comments
    C. Output format
    D. Variable names

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This test questionnaire is for academic purposes only.